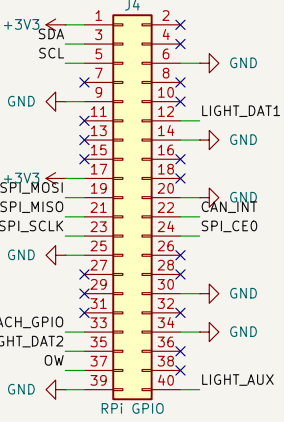
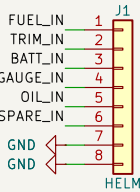


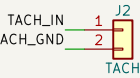
RASPBERRY PI HEADER



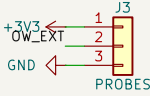
HELM: gauge taps and battery



TACH: Delco EST (Ignition side)

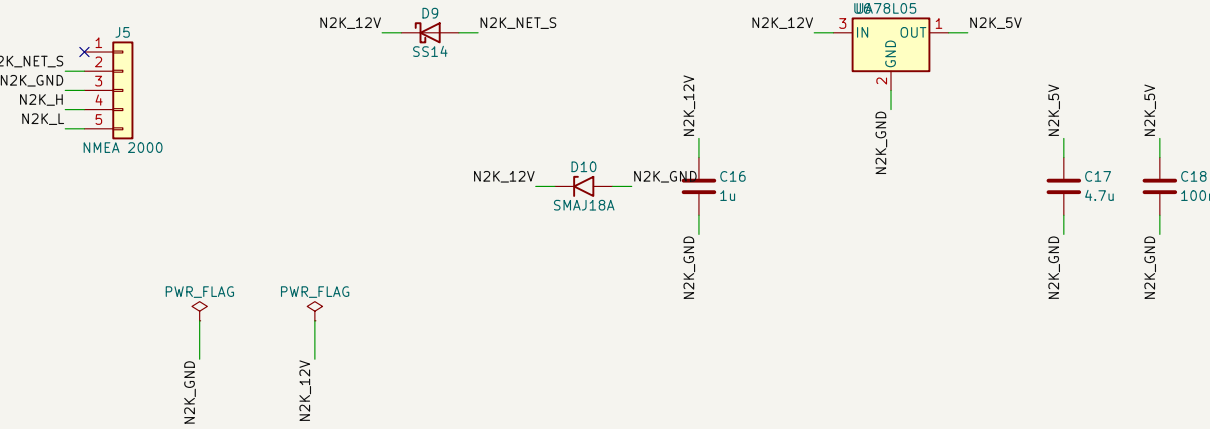


TEMPERATURE PROBES



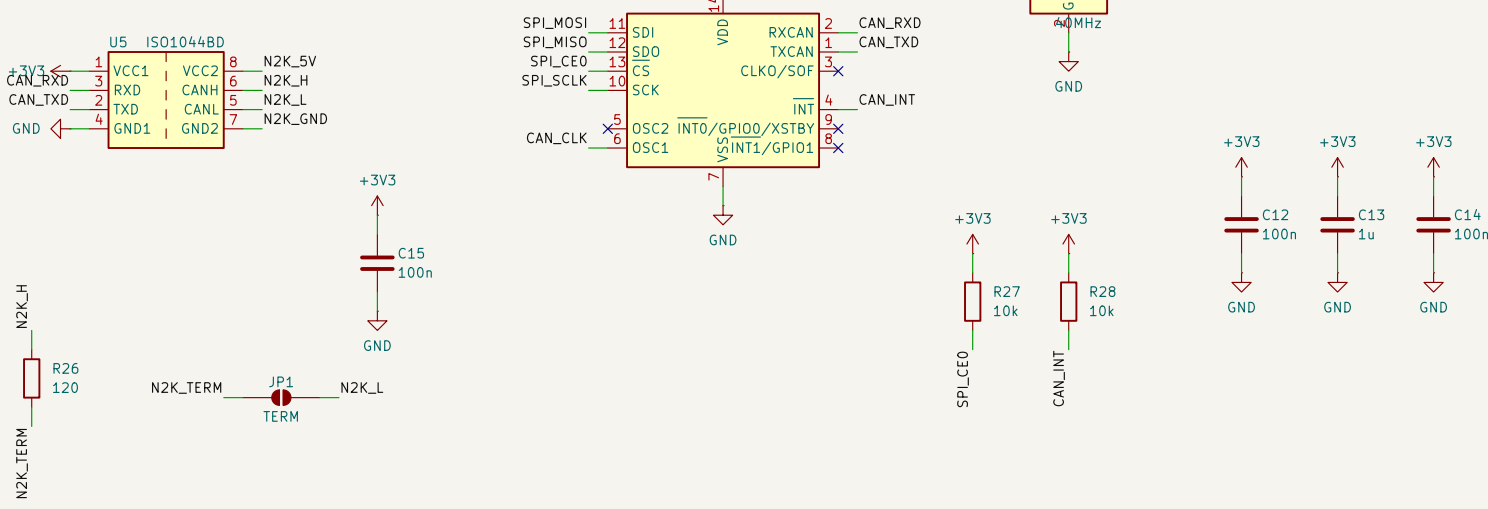
Each gauge tap: the S (or I) terminal -> 10k in heatshink AT THE GAUGE -> J1 -> 39k -> node (10k to GND, BAT54S clamp to GND/3V3) -> 1k -> ADC, with 100n at the ADC pin. V_adc = V_tap x 10/59 (0-16 V -> 0-2.7 V). Battery: 47k on board, no remote resistor. x 10/57. Tach, Delco EST coil (BAT +12 V in, TACH out): J2-1 to the gray wire from the coil's TACH terminal, J2-2 to the tach gauge's G. The tach stays connected. TACH is the coil's switched side -- 12 V, ground while the coil charges, a few hundred volts at each spark: three 3.3k 1206 in series share the spike: 1W4148W and 10n across the LED. One rising edge per ignition cycle on GPIO13; 2 per revolution on the 3.0 4-cylinder (BOAT_TACH_PPR=2). Keep every TACH_* net (the ignition side) 3 mm from all other copper; see sensor-board.kicad_dru. Software: BOAT_SENSORS5=real BOAT_SENDER_WIRING=tap BOAT_TACH_PULL=none BOAT_TACH_GPIO=13 BOAT_OIL_SENDER=true; dtoverlay=w1-gpio,gpiopin=26. Channel map: fuel 0, trim 1, battery 2, gauge supply 3, oil 4 -- see SENSOR_BOARD.md.

NMEA 2000: network side (isolated, powered by NET-S)

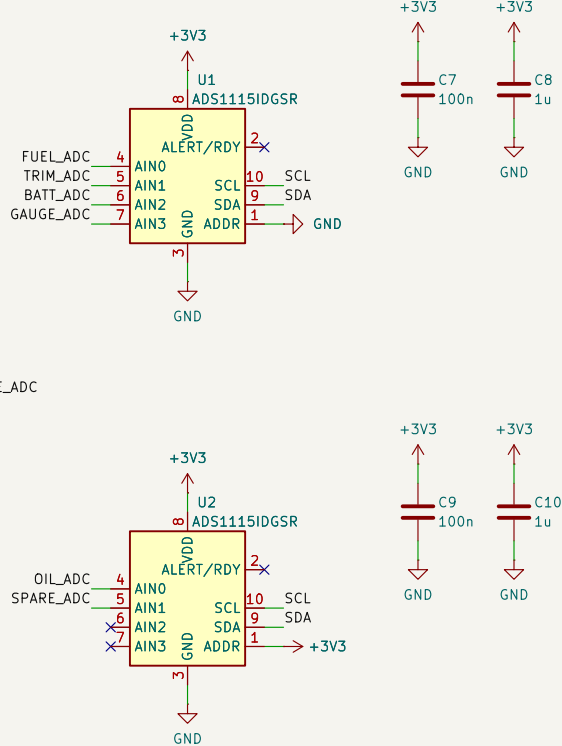


NMEA 2000 (Fusion stereo, depth transducer): J5 takes a drop cable -- 1 shield (not connected), 2 NET-S, 3 NET-C, 4 NET-H, 5 NET-L. U5 isolates the network from the Pi: its network side runs from NET-S through D9 (reverse polarity), D10 (surges) and U6 (5 V), and N2K_GND is the network's 0 V, not the Pi's GND. Keep every N2K_* net 2 mm from all other copper (sensor-board.kicad_dru). About 1 LEN. JP1 (open) adds a 120 ohm terminator for a bench test with no backbone -- never bridge it on the boat; the backbone is terminated at its ends. Pi: dtparam=spi=on dtoverlay=mcp251xf,spi0-0,oscillator=4000000,interrupt=25 -> can0 at 250 kbit/s; BOAT_CAN=can0.

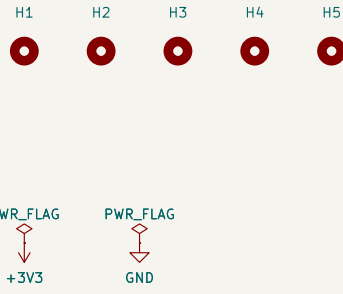
NMEA 2000: PI side



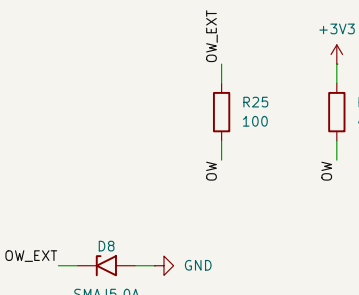
CONVERTERS



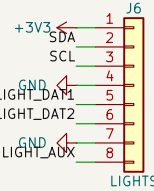
MOUNTING / POWER FLAGS



1-WIRE PROBES



LIGHTS: to a separate LED board



J6 carries logic only: the strips' 12 V and current stay on the LED board. PWM strips: PCA9685s on SDA/SCL (not 0x48/0x49, the converters, or 0x70). Addressable strips: DAT1 = GPIO18 (PWM0), DAT2 = GPIO19 (PWM1), 3.3 V: buffer to 5 V there. AUX = GPIO21, spare. 3V3 for the LED board's logic, under 50 mA.

Generated from design.py by schematic.py; edit those, not this file
Passive taps on the existing analog gauges -- they stay connected and working